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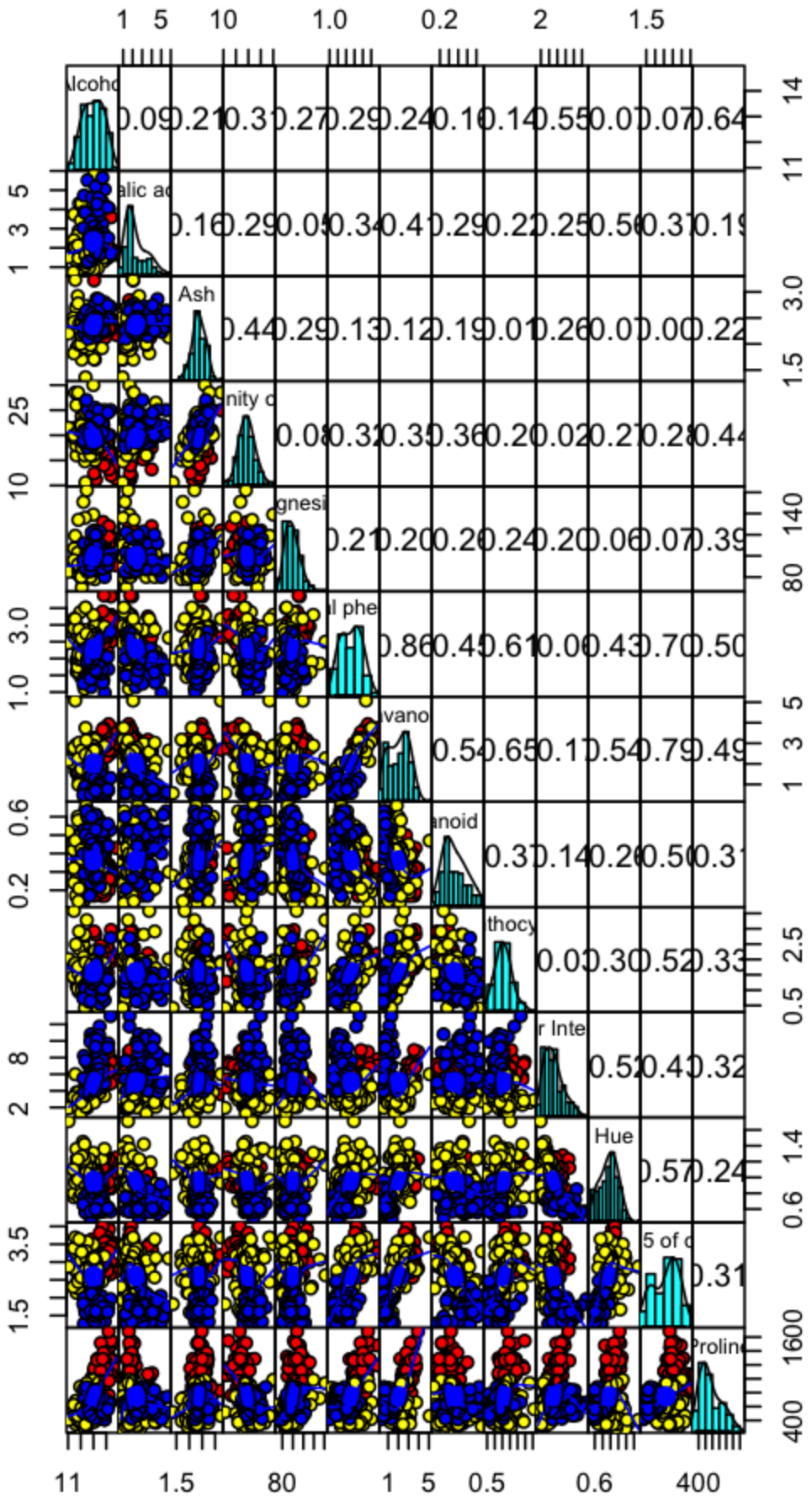
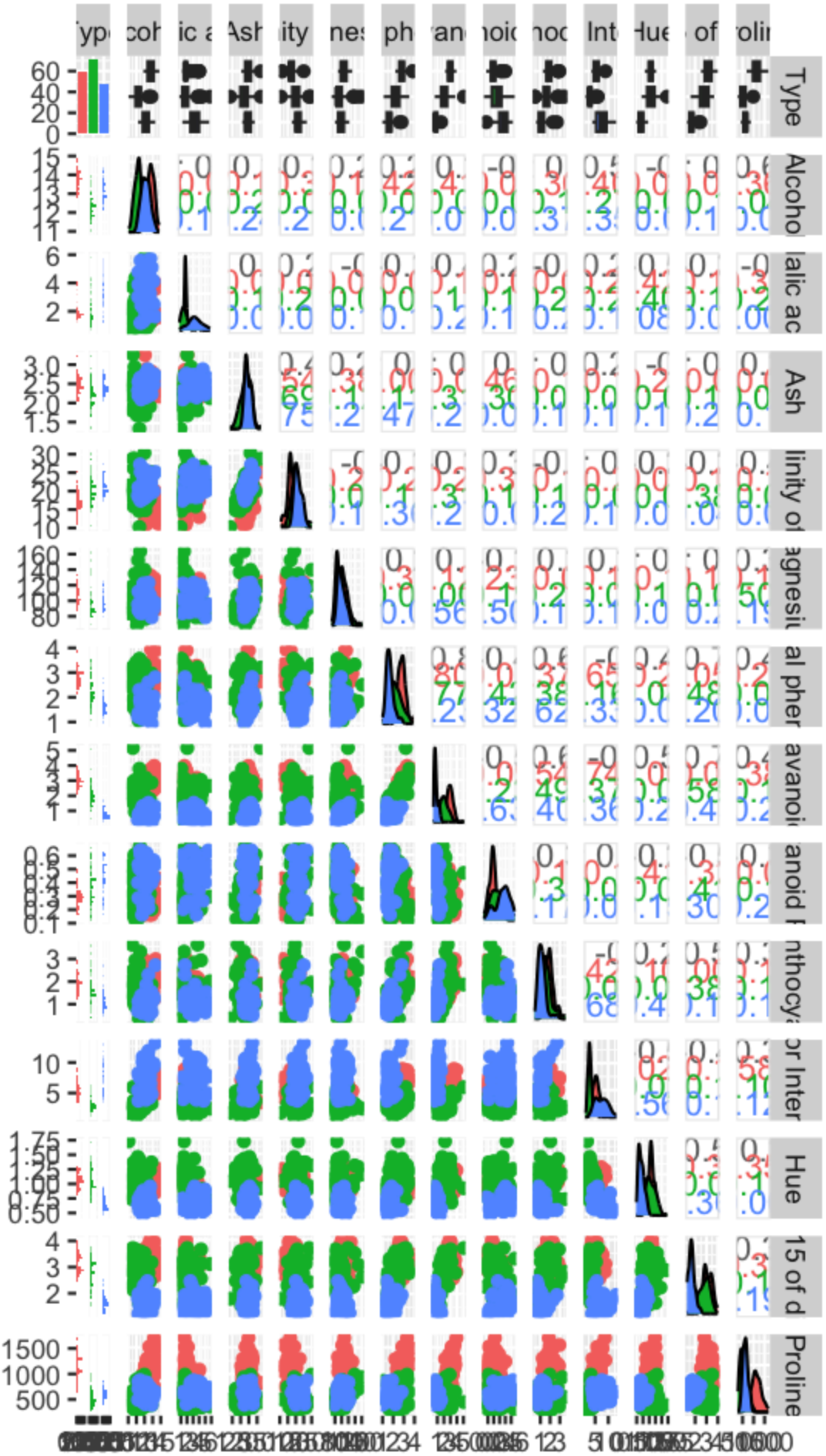
October 24, 2025
Data Analytics Lab 4

```
 ##########################################
> ### Principal Component Analysis (PCA) ###
> ##########################################
>
> ## load libraries
> library(ggplot2)
> library(ggfortify)
> library(GGally)
> library(e1071)
> library(class)
> library(psych)
> library(readr)
>
> ## set working directory so that files can be referenced without the full path
> setwd("~/Downloads")
>
> ## read dataset
> wine <- read_csv("wine.data", col_names = FALSE)
Rows: 178 Columns: 14
── Column specification ────────────────────────────────────────────────────────
────────────────────────────────────────────────────────────────────────────────
Delimiter: ","
dbl (14): X1, X2, X3, X4, X5, X6, X7, X8, X9, X10, X11, X12, X13, X14

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
>
> ## set column names
> names(wine) <- c("Type","Alcohol","Malic acid","Ash","Alcalinity of ash","Magnesium","Total
phenols","Flavanoids","Nonflavanoid Phenols","Proanthocyanins","Color
Intensity","Hue","Od280/od315 of diluted wines","Proline")
>
> ## inspect data frame
> head(wine)
# A tibble: 6 × 14
   Type Alcohol `Malic acid`   Ash `Alcalinity of ash` Magnesium `Total phenols` Flavanoids
`Nonflavanoid Phenols` Proanthocyanins `Color Intensity`   Hue
```

```
  <dbl>   <dbl>       <dbl> <dbl>           <dbl>     <dbl>         <dbl>     <dbl>               <dbl>
<dbl>           <dbl> <dbl>
1     1    14.2        1.71  2.43            15.6       127          2.8       3.06               0.28
2.29            5.64  1.04
2     1    13.2        1.78  2.14            11.2       100          2.65      2.76               0.26
1.28            4.38  1.05
3     1    13.2        2.36  2.67            18.6       101          2.8       3.24               0.3
2.81            5.68  1.03
4     1    14.4        1.95  2.5             16.8       113          3.85      3.49               0.24
2.18            7.8   0.86
5     1    13.2        2.59  2.87            21         118          2.8       2.69               0.39
1.82            4.32  1.04
6     1    14.2        1.76  2.45            15.2       112          3.27      3.39               0.34
1.97            6.75  1.05
# i 2 more variables: `Od280/od315 of diluted wines` <dbl>, Proline <dbl>
>
> ## change the data type of the "Type" column from character to factor
> ####
> # Factors look like regular strings (characters) but with factors R knows
> # that the column is a categorical variable with finite possible values
> # e.g. "Type" in the Wine dataset can only be 1, 2, or 3
> ####
>
> wine$Type <- as.factor(wine$Type)
>
>
> ## visualize variables
> pairs.panels(wine[,-1],gap = 0,bg = c("red", "yellow", "blue")[wine$Type],pch=21)
```

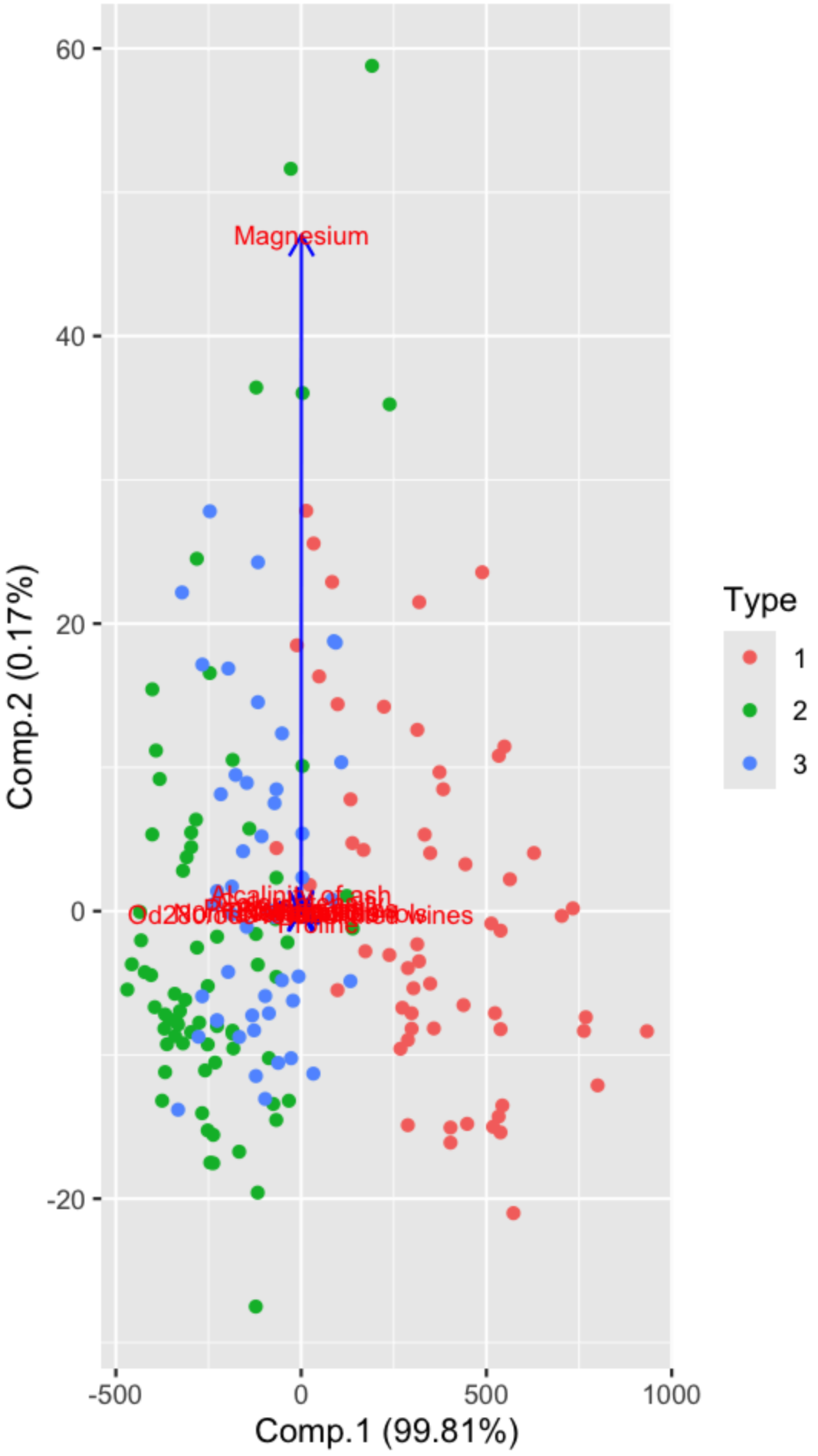

Ash
Hue
Proline

```
>
> ggpairs(wine, ggplot2::aes(colour = Type))
```


Type
Alcoho
lalic ac
Ash
linity of
agnesiu
al pher
avanoi
anoid F
nthocya
or Inter
Hue
15 of d
Proline

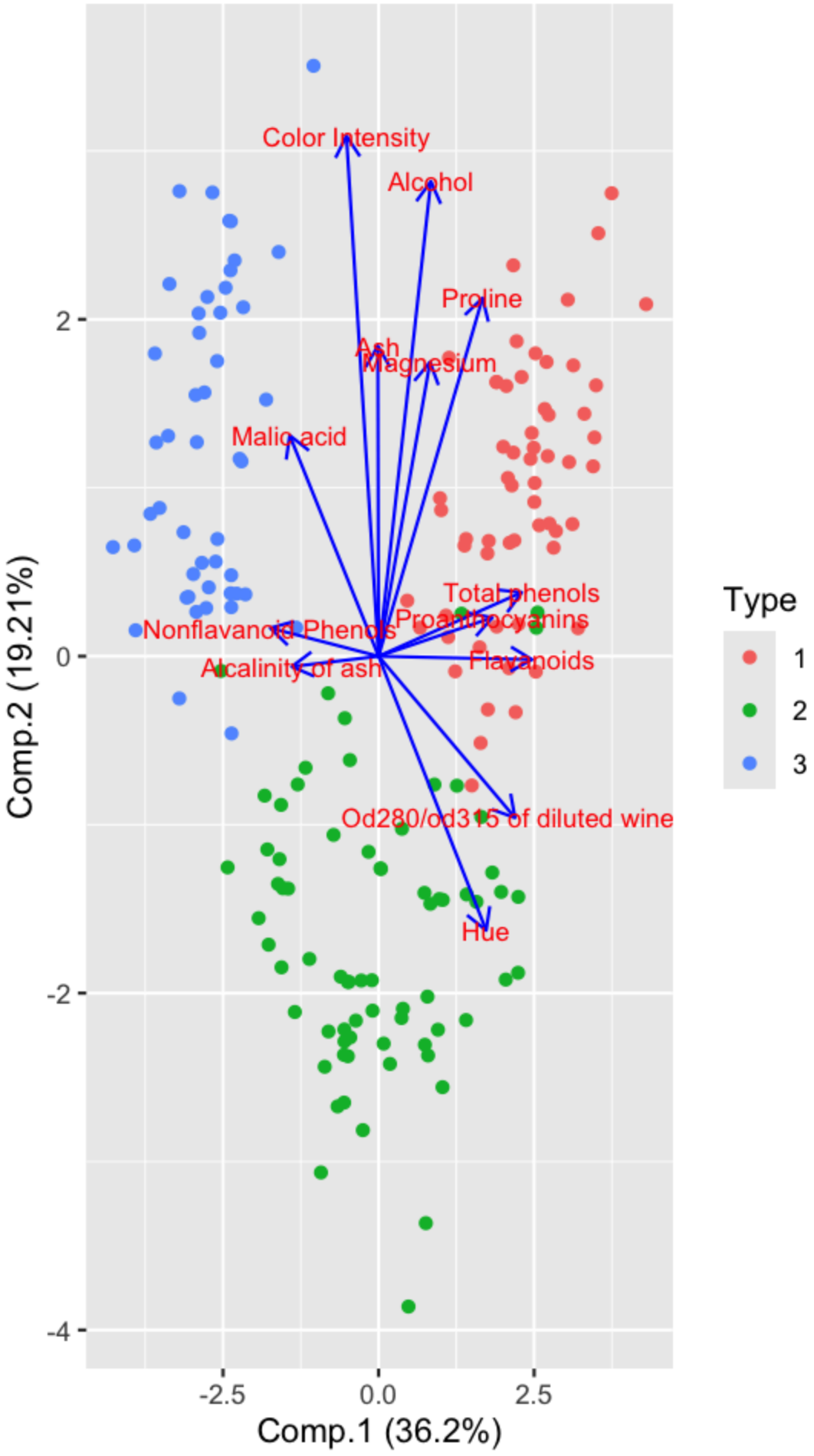
```
>

> ###
> # creating another dataframe from wine dataset that contains the columns from 2 to 14
> X <- wine[,2:14]
> Y <- wine[,1]
>
> ## Compute the PCs and plot the dataset using the 1st and 2nd PCs.
>
> Xmat <- as.matrix(X)
>
> Xc <- scale(Xmat, center = T, scale = F)
>
> principal_components <- princomp(Xc)
>
> summary(principal_components)
Importance of components:
                          Comp.1       Comp.2       Comp.3       Comp.4       Comp.5       Comp.6
Comp.7       Comp.8       Comp.9      Comp.10      Comp.11
Standard deviation     314.0771817 13.098319252 3.063510e+00 2.227810e+00 1.105415e+00
9.145156e-01 5.266937e-01 3.879830e-01 3.338668e-01 2.670202e-01 1.933000e-01
Proportion of Variance   0.9980912  0.001735916 9.495896e-05 5.021736e-05 1.236368e-05
8.462130e-06 2.806815e-06 1.523081e-06 1.127830e-06 7.214158e-07 3.780603e-07
Cumulative Proportion    0.9980912  0.999827146 9.999221e-01 9.999723e-01 9.999847e-01
9.999931e-01 9.999960e-01 9.999975e-01 9.999986e-01 9.999993e-01 9.999997e-01
                          Comp.12      Comp.13
Standard deviation     1.447549e-01 9.031952e-02
Proportion of Variance 2.120138e-07 8.253928e-08
Cumulative Proportion  9.999999e-01 1.000000e+00
>
> principal_components$loadings

Loadings:
                       Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9
Comp.10 Comp.11 Comp.12 Comp.13
Alcohol                                0.141         0.194  0.923  0.285
Malic acid                           0.122  0.160 -0.613  0.742 -0.150
Ash                                                -0.149               -0.954   0.132  -0.174
Alcalinity of ash                    0.939 -0.331
Magnesium                          0.999
Total phenols                                 0.315  0.279        -0.177  0.256 -0.847
Flavanoids                           -0.169  0.525  0.434        -0.248  0.378  0.520   0.133
Nonflavanoid Phenols                                                                  -0.199  -0.148   0.966
Proanthocyanins                               0.251  0.242 -0.310  0.870               -0.136
```

```
Color Intensity                 0.291  0.879  0.332        -0.113
Hue                                                                 -0.976  -0.167
Od280/od315 of diluted wines            -0.178  0.261  0.289  0.102 -0.187 -0.874
Proline                 1.000

               Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9 Comp.10
Comp.11 Comp.12 Comp.13
SS loadings     1.000  1.000  1.000  1.000  1.000  1.000  1.000  1.000  1.000   1.000   1.000
1.000   1.000
Proportion Var  0.077  0.077  0.077  0.077  0.077  0.077  0.077  0.077  0.077   0.077   0.077
0.077   0.077
Cumulative Var  0.077  0.154  0.231  0.308  0.385  0.462  0.538  0.615  0.692   0.769   0.846
0.923   1.000
>
> # using autoplot() function to plot the components
> autoplot(principal_components, data = wine, colour = 'Type',
+          loadings = TRUE, loadings.colour = 'blue',
+          loadings.label = TRUE, loadings.label.size = 3, scale = 0)
```


60
40
20
0
-20
Comp.2 (0.17%)
Magnesium
Alcalinity of ash
Od280/od315 of diluted wines
Proline
-500
0
500
1000
Comp.1 (99.81%)
Type
1
2
3

```
>
> # Following the same steps except with scaled PCA
> Xs <- scale(Xmat, center = T, scale = T)
>
> principal_components2 <- princomp(Xs)
>
> summary(principal_components2)
Importance of components:
                          Comp.1    Comp.2    Comp.3    Comp.4     Comp.5     Comp.6     Comp.7
Comp.8     Comp.9    Comp.10    Comp.11    Comp.12     Comp.13
Standard deviation     2.1631951 1.5757366 1.1991447 0.9559347 0.92110518 0.79878171
0.74022473 0.58867607 0.53596364 0.49949266 0.47383559 0.40966094 0.320619963
Proportion of Variance 0.3619885 0.1920749 0.1112363 0.0706903 0.06563294 0.04935823
0.04238679 0.02680749 0.02222153 0.01930019 0.01736836 0.01298233 0.007952149
Cumulative Proportion  0.3619885 0.5540634 0.6652997 0.7359900 0.80162293 0.85098116
0.89336795 0.92017544 0.94239698 0.96169717 0.97906553 0.99204785 1.000000000
>
> principal_components2$loadings

Loadings:
                               Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9
Comp.10 Comp.11 Comp.12 Comp.13
Alcohol                         0.144  0.484  0.207         0.266  0.214         0.396  0.509  0.212    0.226
0.266
Malic acid                     -0.245  0.225        -0.537         0.537 -0.421               -0.309          -0.122
Ash                             0.316 -0.626  0.214  0.143  0.154  0.149 -0.170 -0.308          0.499
-0.141
Alcalinity of ash          -0.239        -0.612               -0.101  0.287  0.428  0.200         -0.479
Magnesium                       0.142  0.300 -0.131  0.352 -0.727         -0.323 -0.156  0.271
Total phenols                   0.395         -0.146 -0.198  0.149               -0.406  0.286 -0.320  -0.304
0.304  -0.464
Flavanoids                      0.423         -0.151 -0.152  0.109               -0.187         -0.163
0.832
Nonflavanoid Phenols          -0.299         -0.170  0.203  0.501 -0.259 -0.595 -0.233  0.196  0.216
-0.117          0.114
Proanthocyanins                 0.313         -0.149 -0.399 -0.137 -0.534 -0.372  0.368 -0.209  0.134
0.237          -0.117
Color Intensity                 0.530  0.137               -0.419  0.228               -0.291          -0.604
Hue                          0.297 -0.279         0.428  0.174  0.106 -0.232  0.437         -0.522
-0.259
Od280/od315 of diluted wines  0.376 -0.164 -0.166 -0.184  0.101  0.266               0.137  0.524
-0.601  -0.157
Proline                      0.287  0.365  0.127  0.232  0.158  0.120         0.120 -0.576  0.162  -0.539
```

```
                Comp.1 Comp.2 Comp.3 Comp.4 Comp.5 Comp.6 Comp.7 Comp.8 Comp.9 Comp.10
Comp.11 Comp.12 Comp.13
SS loadings     1.000  1.000  1.000  1.000  1.000  1.000  1.000  1.000  1.000   1.000   1.000
1.000   1.000
Proportion Var  0.077  0.077  0.077  0.077  0.077  0.077  0.077  0.077  0.077   0.077   0.077
0.077   0.077
Cumulative Var  0.077  0.154  0.231  0.308  0.385  0.462  0.538  0.615  0.692   0.769   0.846
0.923   1.000
>
> library(caret)
>
> # using autoplot() function to plot the components
> autoplot(principal_components2, data = wine, colour = 'Type',
+          loadings = TRUE, loadings.colour = 'blue',
+          loadings.label = TRUE, loadings.label.size = 3, scale = 0)
```


Color Intensity
Alcohol
Proline
Ash
Magnesium
Malic acid
Total phenols
Nonflavanoid Phenols
Proanthocyanins
Alcalinity of ash
Flavanoids
Od280/od315 of diluted wine
Hue
Type
1
2
3
2
0
-2
-4
-2.5
0.0
2.5
Comp.2 (19.21%)
Comp.1 (36.2%)

```
> ## Identify the variables that contribute the most to the 1st PC and 2nd PCs.
> # Color intensity, alcohol, proline, total phenols, flavanoids, Od280/od315 of diluted wines, and hue contribute the most
>
> ## Train a classifier model (e.g. kNN) to predict the wine type using all the variables in the original dataset.
> split.rat <- 0.7
> train.indexes <- sample(150,split.rat*150)
>
> train <- wine[train.indexes,]
> test <- wine[-train.indexes,]
>
> mod.knn <- train(Type~., data=train, method="knn")
>
> ### 1 round cross-validation
> knn.train.true <- train[,1]
> knn.test.true <- test[,1]
>
> knn.train.predicted <- predict(mod.knn,train[,-1])
> knn.test.predicted <- predict(mod.knn,test[,-1])
>
> train.cm = as.matrix(table(Actual = knn.train.true$Type, Predicted = knn.train.predicted))
> train.cm
      Predicted
Actual  1  2  3
     1 39  2  0
     2  5 45  3
     3  3  8  0
> train.accuracy <- sum(diag(train.cm))/nrow(train)
> train.accuracy
[1] 0.8
>
> test.cm = as.matrix(table(Actual = knn.test.true$Type, Predicted = knn.test.predicted))
> test.cm
      Predicted
Actual  1  2  3
     1 17  1  0
     2  0 17  1
     3  6 30  1
> test.accuracy <- sum(diag(test.cm))/nrow(test)
> test.accuracy
[1] 0.4794521
>
```

```
> ##Train a classifier model to predict the wine type using the data projected onto the first 2 PCs
(scores in the princomp function's return object)
> pc_scores <- as.data.frame(principal_components2$scores[, 1:2])
> colnames(pc_scores) <- c("PC1","PC2")
> pc_data <- data.frame(Type = as.factor(wine$Type), pc_scores)
>
> train.pc <- pc_data[train.indexes, ]
> test.pc  <- pc_data[-train.indexes, ]
>
> mod.knn <- train(Type ~ PC1 + PC2, data = train.pc, method = "knn")
>
> knn.train.true <- train.pc[, 1]; knn.test.true <- test.pc[, 1]
> knn.train.predicted <- predict(mod.knn, train.pc[, -1])
> knn.test.predicted  <- predict(mod.knn,  test.pc[, -1])
>
> train.cm <- as.matrix(table(Actual = knn.train.true, Predicted = knn.train.predicted))
> test.cm  <- as.matrix(table(Actual = knn.test.true,  Predicted = knn.test.predicted))
>
> ## Compare the 2 classification models using contingency tables and precision/recall/F1
metrics.
> train.cm <- as.matrix(table(Actual = train.pc$Type, Predicted = knn.train.predicted))
> train.cm
      Predicted
Actual  1  2  3
     1 40  1  0
     2  3 49  1
     3  0  0 11
> train.accuracy <- sum(diag(train.cm)) / nrow(train.pc)
> train.accuracy
[1] 0.952381
>
> test.cm <- as.matrix(table(Actual = test.pc$Type, Predicted = knn.test.predicted))
> test.cm
      Predicted
Actual  1  2  3
     1 18  0  0
     2  0 18  0
     3  0  1 36
> test.accuracy <- sum(diag(test.cm)) / nrow(test.pc)
> test.accuracy
[1] 0.9863014
>
> knn.cm.full <- confusionMatrix(knn.test.predicted, test.pc$Type, mode = "prec_recall")
> knn.byclass <- as.data.frame(knn.cm.full$byClass)
```

```
>
> knn.precision_macro <- mean(knn.byclass$Precision)
> knn.recall_macro    <- mean(knn.byclass$Recall)
> knn.f1_macro        <- mean(knn.byclass$F1)
>
> knn.precision_macro
[1] 0.9824561
> knn.recall_macro
[1] 0.990991
> knn.f1_macro
[1] 0.9864248
```